

Vision Banana: Evidence, Capabilities and a Feasible Research Contribution

ELEC4240 research briefing | Based on the linked paper v1 | 2 October 2026

HO, Chun Wai (21053878) / Tong, Man Hung (21064669) / MA, Shenhan (21041382)

Main assessment. Vision Banana provides experimental evidence that an image generator can be adapted into a unified model for several perception tasks. For our project, the most practical research direction is to investigate when geometric consistency actually improves depth and normal prediction.

What does the paper establish?

The model instruction-tunes Nano Banana Pro using a mixture of its original data and added vision-task data. Different prompts select different tasks while sharing the same model weights. Its central design represents task outputs as RGB images that can be decoded into usable predictions. On the reported benchmarks, it matches or surpasses several specialist models and broadly retains image generation and editing capabilities. These results support the transferability of representations learned through generative pretraining. [1]

Input image + task instruction > Image generator > RGB output > Task-specific decoding

What tasks can it perform?

Task	Output and practical meaning
Semantic segmentation	Assign a category, such as road, building or person, to each pixel.
Instance segmentation	Separate individual objects that may belong to the same category.
Referring-expression segmentation	Find an object described in language, such as a person in a pink shirt.
Metric depth estimation	Predict per-pixel depth in metres from one image.
Surface normal estimation	Predict the orientation of visible surfaces.
Image generation and editing	Continue generating images and editing their content.

The project also visualizes point clouds obtained by combining predicted depth with camera intrinsics. This is a derived use of depth, rather than evidence of complete 3D reconstruction from any image. [2]

Evidence boundary. “Zero-shot transfer” still involves vision-task instruction tuning; the benchmark training splits are excluded from that tuning mixture. The model is not best on every dataset. Metric output does not guarantee exact measurement on arbitrary photos, and the results do not establish equivalent performance with a small model, little data or limited compute. [1]

A research question grounded in our results

Our Marigold implementation is an open-model study inspired by Vision Banana. Differences in backbone, data and training scale mean that it is neither a complete reproduction nor a direct test that could refute the original paper.

Proposed question. Under limited data and parameter-efficient adaptation, can the reliability of geometric supervision reduce negative transfer between metric depth and surface normal prediction?

What our completed experiment shows

V7 setting	Depth AbsRel	Normal error
Depth only	0.37068	Not applicable
Normals only	Not applicable	39.83°
Shared joint adaptation	0.41819	42.08°
Joint adaptation + geometry	0.39685	45.07°

Lower is better. Values are averages across three seeds, using 128 training scenes, 32 previously observed validation scenes and 320 updates per run. Normal errors are measured against normals derived from depth, not independent normal ground truth. [3]

Adding geometry reduced prediction-to-prediction angular disagreement from **48.54° to 31.11°**, while normal-target error rose from **42.08° to 45.07°**. The mean depth improvement over ordinary joint adaptation did not pass the exploratory Holm adjustment ($p = 0.19344$); normal degradation did ($p = 0.00020$). Thus, greater agreement can coexist with worse target accuracy. [3]

This motivates an investigation, but does not isolate its cause. Label sensitivity, short training, shared adapter capacity and optimization remain competing explanations.

Candidate modification: reliability-weighted geometry

- 1. Estimate label stability.** Derive normal labels from each training depth map using several fixed neighbourhood sizes or smoothing scales. Measure angular disagreement at each valid location.
- 2. Weight the geometric term.** Assign higher weights to stable locations and lower weights to unstable ones. Retain both original supervised objectives and the existing validity mask.
- 3. Validate the proxy first.** Use known planes, corners, controlled noise and missing depth to test whether instability predicts actual normal error. Stable labels can still be wrong.
- 4. Keep inference unchanged.** Compute weights from training labels and detach them from optimization. Ground-truth depth and reliability maps are not required at inference.

Status: this candidate weighting method has not been implemented, trained or evaluated. No improvement is claimed.

How to establish a credible contribution

Minimum informative ablation

Condition	Question it answers
Joint model without geometry	What is the shared-model baseline?
Uniform geometric penalty	What does the existing constraint change?
Weaker uniform penalty	Is a gain explained simply by less regularization?
Reliability-weighted penalty	Does the proposed weighting rule help?
Spatially shuffled weights	Does selecting the correct locations matter?

Retain matching depth-only and normal-only controls. Match training data, initialization, exposure, resolution and checkpoint selection. Report all seeds, both task errors and compute cost. Keep the evaluation mask common to every method.

Use current scenes for development. Freeze the selected method and statistical comparisons before testing genuinely unobserved scenes. To claim no meaningful normal degradation, predefine an acceptable margin and evaluate the corresponding confidence bound; a non-significant difference alone is insufficient.

Where the novelty must be precise

Joint depth-normal diffusion is already studied by GeoWizard [4]. Uncertainty-weighted geometric consistency also has precedents: Li et al. use local point-to-plane residuals to reduce depth-to-normal conversion bias [5]. Therefore, joint prediction, geometric consistency and uncertainty weighting are not new in themselves.

Potential contribution: a specific spatial weighting rule based on the sensitivity of derived labels to their construction scale, together with controlled evidence of when it helps or fails during low-cost generative-model adaptation. This is a candidate course-project contribution; publication-level novelty needs a broader comparison with prior work.

Recommended sequence: stabilize single-task training, validate the reliability proxy, run the controlled ablations, and then evaluate the frozen method independently.

Suggested working title: Reliability-Aware Geometric Supervision for Parameter-Efficient Depth and Normal Adaptation

Sources

- [1] [Vision Banana, paper v1](#)
- [2] [Vision Banana, official project demonstrations](#)
- [3] [Project method plan and completed V7 evidence](#)
- [4] [GeoWizard, ECCV 2024](#)
- [5] [Li et al., geometric constraints for laparoscopic depth, MICCAI 2025](#)